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The vehicle routing problem in field logistics: Part II

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Most agricultural field operations involve a number of highly interconnected tasks executed by co-operating heterogeneous agricultural machines. Multiple machinery systems involved in "output material flow" operations, such as harvesting, as well as in "input material flow" operations, such as spraying and fertilising, include a number of primary units (PUs) supported by a number of service (mainly transport) units (SUs). Such operations require considerable efforts in terms of the managerial tasks of scheduling and planning. Here, a theoretical approach is presented to represent the planning and scheduling tasks for the SUs as examples of the well-known combinatorial optimisation problem "vehicle routing problem with time windows" (VRPTW). The approach builds on the concept that in the case of field operations involving co-operating machines, an SU, or a team of SUs, is required fulfil a request for on-site services from PUs, where the requests are generated by a spatial–temporal process which may be deterministic (*e.g.*, seeding), stochastic (*e.g.*, harvesting) or dynamic (*e.g.*, sensor-based site-specific spraying). According to this concept, the PU can be considered as the "customers" in the vehicle routing problem methodology.

It has been shown that scheduling and planning problems for SUs in conventional machinery systems, as well as in emerging field robot systems, can be cast as examples of the VRPTW and, consequently, can be solved using advanced methods that have developed for the solution of these examples.

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1. Introduction

Most of the agricultural field operations (AFOs) involve a number of highly interconnected tasks executed by co-operating heterogeneous agricultural machines. For example, this is the case for operations that take place within the biomass supply chain such as harvesting, material handling, removal of biomass from the field, and rural and public road transportation. This machinery system includes, between each pair of successive stages in the chain, a number of heterogeneous machines, *e.g.*, harvesters, transport units, medium and high-capacity transport trucks, and unloading and pre-processing equipment. Multiple machinery operations of similar planning complexity as harvesting ("output material flow – OMF") are operations such as spraying and fertilising ("input material flow – IMF"). These operations also involve multi-machinery systems that potentially include one or a number of self-propelled or tractor-pulled application units and one or more transport units (Sørensen and Bochtis, 2010).

Two main managerial tasks have to be considered in these kinds of operations: scheduling and operational planning. Scheduling has to do with the allocation and commitment of recourses (machines, labour, materials) to fields and the

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Nomenclature

at_i^k	the adjusted time needed for the starting of the service procedure between the s_k SU and the p_i PU or f_i FU
a_i	the starting time of the time window wherein customer $i \in N$ has to be served
b_i	the ending time f the time window wherein customer $i \in N$ has to be served
E	the set of arcs (edges) of the graph G: $E = N \times N$
ε_i^k	the service times associated with each node in the case of the time windows presence ($i \in P \wedge F, k \in S$)
C_{index}	the capacity of a mobile unit: $index \equiv i \in V$, in general the capacity of a vehicle as in VRP, $index \equiv p_k, k \in P$, capacity of a PU, $index \equiv s_k, k \in S$, capacity of a SU
c_{ij}	the cost associated with each arc $(i, j) \in E$ representing the transit cost from i to j
d_i	the non-negative demand that is associated with each customer $i \in N$
F	the set of the FU, each referenced to as f_k, $k \in F$
G	the complete graph $G = \{N, E\}$
L	the set of the different machine types
N	the set of nodes in the VRP graph
P	the set of the PU, each referenced to as p_k, $k \in P$
R_j	the permutation set that includes the ordering of the nodes visited by vehicle v_j, $j \in V$ according to the VRP solution, $R_j = \langle i_1, i_2, \ldots, i_{\lvert R_j \rvert} \rangle$
S	the set of the SU, each referenced to as s_k, $k \in S$
SR_{index}	the service rate; if $index \equiv p_i, i \in P$ service rate corresponds to the unloading rate of the harvester's auger, if $index \equiv s_i, i \in S$ service rate corresponds to the refilling rate of the SU in the material input operation type, if $index \equiv f_i, i \in F$ service time corresponds to the refilling or unloading rate (depending on the operation type) of the facility's equipment
sc_k	the number of the service calls expected by PU p_k
t_{ij}	the travel time associated with each arc $(i, j) \in E$
V	the set of the vehicles available at the depot, each referenced to as v_k, $k \in V$
γ	the penalty cost added when latest time of a time window is violated
τ_{ij}	the total time the SU, s_i, $i \in S$, spends before reaches the PU p_j, $j \in P$

Abbreviations

AFO	agricultural field operation
CTF	controlled traffic farming
DVRPTW	dynamic vehicle routing problem with time windows
IMF	input material flow
FCFS	first-come first-served (queuing system rule)
FU	facility unit
GIS	geographical information system
MDVRPTW	multiple depot vehicle routing problem with time windows
mod TSP	modified travelling salesman problem (strategy)
OMF	output material flow
PU	primary unit
SU	service unit
SVRPTW	stochastic vehicle routing problem with time windows
UCTF	un-controlled traffic farming
VRP	vehicle routing problem
VRPTW	vehicle routing problem with time windows
VRPTWSD	vehicle routing problem with time windows and stochastic demands

assignment of service units (SUs) (*e.g.*, transport carts, and refilling units) to primary machinery units (*e.g.*, combines, application units) taking into account the time horizon. On the other hand, operational planning has to do with the on-line coordination of the operating system's components. For example, in the case of harvesting, on-line decisions concern which transport cart will unload which combine, in what sequence, which silo should be used for which unloading of the transport carts, *etc.* Scheduling and operational planning of farm operations involves stochastic optimisation to deal with the time and space variability caused by the interaction between biological, metrological and machinery factors.

A standard method used in agricultural machinery management for evaluating systems with interacting machines or components is cycle analysis (Hunt, 2001; Buckmaster, 2006). This method can be used to estimate all time elements spent by each machine in the system and can be useful to identify suitable transport systems in field operations. Another method involves non-linear programming optimisation model based on a level of aggregation consistent with the accessible and existing data related to machinery sets, crops, weather and timeliness of operations (Søgaard and Sørensen, 2004). Recently, tools that take into account time evolution have been implemented. Busato *et al.* (2007) developed a dynamic discrete event simulation model in order to optimise the wheat harvesting and transport operations accounting for field size and shape, field distance to silo, yield, and available resources. An enhanced version of this system was presented by Berruto and Busato (2008) in which the event-oriented simulation is combined with linear programming for the evaluation of the biomass supply chain. In general, the scheduling of large scale harvesting operations significantly differs from the application of scheduling models currently available in the literature. This is caused by the differences in the interchangeability of the workers and equipment employed to carry out the operations, the need to schedule each of these resources individually, and the presence of time lags (Basnet *et al.*, 2006). Based on the fact that agricultural operations involve both continuous and discrete events, Guan *et al.* (2008) introduced the hybrid Petri nets method as a modelling tool for farm work flows in agricultural production. In their approach, the continuous part of the hybrid Petri nets method mainly models the practical work in the farmland, while the discrete part mainly represents the status changes in resources such as machinery and labour, as well as, present status or unwanted interruptions during the

farming process. The simulated results revealed that the hybrid Petri nets model was promising as a method that can describe the farming process and reallocate resources in the presence of uncertainties.

From the previous research, it is evident that the scheduling of the operations is not related to the operational planning or to the on-line coordination of the heterogeneous agricultural machines. These managerial tasks are becoming increasingly important as integral parts of the emerging technology of autonomous agricultural machines. According to the current vision for autonomous machines, large teams of smaller and simpler autonomous machines are to replace smaller groups of heavier and complex machines as a way to build collective behavioural systems that are more economical, more scalable and less susceptible to overall malfunction (Blackmore *et al.*, 2007). The operational planning of these teams should be tackled concurrently with other issues being investigated such as communication architecture, coverage planning (Oksanen and Visala, 2007; Bochtis and Vougioukas, 2008), path planning (Linker and Blass, 2008; Bochtis *et al.*, 2009a,b), heterogeneous robots docking (Hao *et al.*, 2004), and reaction control, and multi-robot motion coordination dealing with unsafe conditions and unexpected situations caused by obstacles (Van Henten *et al.*, 2003) or by the presence of other robots (Vougioukas *et al.*, 2006).

Here, a planning approach for the operations of auxiliary agricultural machines is presented, based on the conceptual application of the well-known combinatorial optimisation problem vehicle routing problem (VRP), and specifically of its class that includes time windows constraints (vehicle routing problem with time windows – VRPTW). This paper constitutes a continuation of the work presented by Bochtis and Sørensen (2009) on the conceptualisation of AFOs as instances of VRPs. According to that abstraction, the tracks worked in the field was represented as the ''customers'' in the VRP methodology. According to the concept presented in this paper, the supported primary machines are the ''customers'' in the VRP methodology. The approach was motivated by the fact that in agricultural operations involving co-operating machines, a number of SUs are required to fulfil requests for on-site service of a number of primary units (PUs) in a given field region and at a specific time. Furthermore, service requests are generated by a spatial–temporal process which may be deterministic (*e.g.*, seeding), stochastic (*e.g.*, harvesting) or dynamic (*e.g.*, sensor-based site-specific spraying). All the previous planning aspects have been successfully addressed in various instances of the VRPTW.

2. Preliminaries

2.1. The VRPTW

2.1.1. General

The VRPTW is a generalisation of the classic VRP. The VRP can be described as the problem of determining least cost routes in terms of an optimisation criterion (usually the travelled distance) for a fleet of vehicles from one depot to a set of geographically dispersed points corresponding to some type of ''customers'' (cities, stores, warehouses, schools, *etc*). The routes must be generated in such a way that each customer is visited only once by exactly one vehicle, all routes start and end at the depot, and the total demands of all customers on one particular route will not exceed the capacity of the vehicle. The VRPTW involves the added complexity that every customer should be served within a given time window. Specific examples of problems with time windows include bank or postal deliveries, industrial refuse collection, school-bus routing, grocery distribution, dial-a-ride services, repairmen scheduling, and airline and railway fleet routing and scheduling.

The VRPTW, being the subject of intensive research efforts has generated significant research interest over the years resulting in major contributions in the area of combinatorial optimisation. Because of the difficulty of the VRPTW and its wide applicability to real-life situations, many heuristic solution techniques capable of producing high-quality solutions in limited time have been proposed; a recent thorough survey on the heuristics used for the VRPTW can be found in Bräysy and Gendreau (2005a,b). In real-life applications, requests for service often arrive sequentially in time, and these arrival epochs may be stochastic; moreover, locations of future demands may be unknown or known only probabilistically. In order to meet the demands from multiple real world routing problems which in practice are inherently dynamic and stochastic, a number of advanced instances of the static and deterministic classic VRPTW have been developed (Cordeau *et al.*, 2000).

2.1.2. Notation

Given a number n of customers and a set V representing a fleet of homogenous vehicles (each referent to as v_k, $k \in V$), the basic form of the VRPTW is related to a graph $G = \{N, E\}$, where $N = \{0,1,\ldots,n+1\}$ is the node set and E is the arc set. Nodes 0 and $n+1$ corresponds to the vehicles depot (the ''driving-out'' and the ''returning'' depot, respectively), while the remaining node set $N\backslash\{0, n+1\}$ corresponds to the customers. At each node (including the one corresponding to the depot nodes) a service time ε_i is associated denoting the period of time a vehicle spends in order to service the customer and a time window $[a_i, b_i]$ wherein the customer has to be served. A non-negative cost c_{ij} representing the transit cost from i to j and a travel time t_{ij} are associated with each arc $(i, j) \in E$.

Each vehicle, v_k, $k \in V$, has a capacity C and each customer, $i \in N/\{0, n+1\}$, is associated with a non-negative demand of d_i units ($d_0 = d_{n+1} = 0$).

The solution to the VRPTW consist of determining one route for each vehicle ($|V|$ permutation sets of the form $R_j = \langle i_1, i_2, \ldots, i_{|R_j|} \rangle$), with minimum cost, satisfying the following constraints:

(i) Each route originates at the ''driving-out'' depot and ends at the ''returning'' depot *i.e.*, ($i_1 = 0$, $i_{|R_j|} = n+1$ and $\{i_2, \ldots, i_{|R_j|-1}\} \subseteq N \setminus \{0, n+1\}$).
(ii) Each customer is serviced exactly once, *i.e.*, $\cap_{j=1}^{|V|} R_j = \{0\} \wedge \cup_{j=1}^{|V|} R_j = N$.
(iii) The total demand of all costumers visited in a route does not exceed the vehicle capacity, *i.e.*, $\sum_{h=2}^{|R_j|-1} d_{i_h} \leq C$.
(iv) The time windows constraints are observed, *i.e.*, $a_{i_{k-1}} + \varepsilon_{i_{k-1}} + t_{i_{k-1} i_k} \leq b_{i_k}$, where $i_{k-1}$and i_k represent

Table 1 – Machinery classification

Type	Number	Team identity
PU	One (1)	Identical (≡)
SU	Multiple (m)	Non-identical (≠ {kin, cap})

successive costumers within a route. Furthermore, feasible solutions exist only if $a_0 \leq \min_{i \in N \setminus \{0+1\}} (b_i - t_{0i})$ and $b_{n+1} \geq \min_{i \in N \setminus \{0\}} (a_i + \varepsilon_i + t_{i0})$.

Finally, it must be noted that when vehicles are allowed to remain at the depot, the arc $(0, n+1)$ with $c_{0,n+1} = t_{0,n+1} = 0$ must be added to the arc set A. This complies with the case where the primary objective consists of minimising the number of vehicles used.

2.2. *Agricultural field operations – notation*

AFOs which incorporate the notion of time windows are operations executed by heterogeneous cooperative machines. The involved machines in this type of operations can be categorised in two general types: PUs that perform the main work task (*e.g.*, harvesting, fertilising, spraying, and seeding) and SUs that support the PUs, for example, the transport carts being used for the out-of-the field removal of the harvested grains in a harvesting operation or the carts being used to carry the fertiliser for the refilling of an application unit in an fertilising operation (Bochtis and Sørensen, 2009). Let P denotes the set of the PU, each referent to as p_k, $k \in P$, and S denotes the set of the SU, each referenced to as s_k, $k \in S$. This paper focus on the optimal planning for the SUs ($S \equiv V$), while the PUs constitute the "customers" of the planning problem (as it will be seen later, one PU represents a number of "customers"). Both sets of PUs and SUs can be composed by one or a number of machines which can be identical or non-identical (Table 1). The latter regards the case in which units have different kinematic features and/or different carrying volume capacities. In the case of AFOs, the SUs use facility units (FUs) to unload the material that is carried out-of-the field or to reload with the material that is carried into the field. These FUs corresponds to the depots in the VRP. Let F denotes the set of the FUs, each referenced to as f_k, $k \in F$. For generalisation purposes, it is assumed that there is one SU assigned to each FU, since the case where more than one SU is assigned to a FU can directly be applied to the same assumption. Indeed, considering the subsets $S_k \subseteq S, 1 \leq k \leq |F|$, of the assigned SUs to each FU, is equivalent to the case in which there are $|F'| = \sum_{|F|} S_k$ FUs and each FU has exactly one SU. A FU can be stationary (*e.g.*, a silo or the farm depot) or mobile (*e.g.*, a transport truck) (Table 2).

AFOs that are executed by cooperative machines involve a "flow" of a material. If a quantity of a "commodity" is transported by a SU and is distributed in the field area by a PU, the corresponding operation can be referred to as IMF operation (*e.g.*, seeding, fertilising, and spraying). Equivalently, if a quantity of a "commodity" is transported out-of-the field area by a SU, the corresponding operation can be referred to as OMF operation (*e.g.*, harvesting). Depending on the available information regarding the quantity of the "commodity" that has to be removed or distributed from/in the field, three types of demand characteristics can be distinguished (Table 3):

Table 2 – Facilities classification

Number	Type
One (1)	Stationary (sta)
Multiple (m)	Mobile (mob)

Table 3 – Operations classification

Material flow	Demand nature
IMF	Deterministic (+)
OMF	Stochastic (~)
	Unknown (!)

Deterministic (+). This is the case of *a priori* knowledge available of demand, for example, in a seeding operation. The seed quantity per field area unit is predetermined.
Stochastic (~). In this case, there is no precise knowledge of demands, but it is expected to follow a known probability distribution, as for example in the case the predicted yield distribution.
Unknown (!). In this case, the actual demand is unknown and there is no probabilistic knowledge of it available, as for example in the case of the operation of real-time spraying based on sensor data.

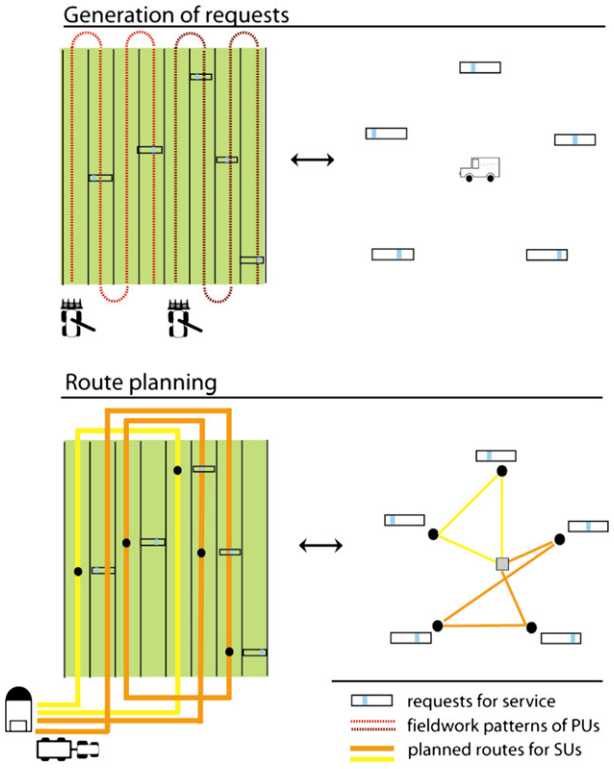


Fig. 1 – Correspondence between SUs field operations and VRPTW.

3. Casting agricultural field operations into examples of the vehicle routing problem with time windows

3.1. Graphical representation

According to the denoted representation, the customers (*i.e.*, the nodes of the graph) correspond to expected requests for on-site service from the PUs (Fig. 1). Consequently, the identity of each node is determined by the estimated location where the corresponding PU should be serviced (refilling or unloading will take place), the PU's specifications (capacity, service time-loading or reloading time), and the time window within which rendering of service should take place. Let sc_k denote the number of the service calls expected by PU p_k.

The assigned demand, d_i, to each node (service location) is equal to the tank capacity, C_{pk}, $k \in P$, of the PU that has to be serviced at the corresponding location. A time window, $[a_i, b_i]$, $i \in N$, is associated with each node in $N \setminus \{0, 1+\sum_{i=1}^{|P|} sc_k\}$, which imposes the restriction that a SU must arrive at the service location beforeb_i.

As mentioned in Section 2.2, at each node (including the nodes corresponding to the depot) a service time has to be associated denoting the period of time the SU spends in order to service the PU (or to be served at the FU in the case of the depot nodes). In the general case, where the team of the SUs composed by heterogeneous machines, for each PU and FU a total number of $|S|$ service times $\varepsilon_i^k, i \in P \wedge F, k \in S$, should be determined by the following expression building on the principles of decomposition of the task into time elements (*e.g.*, Sørensen *et al.*, 2003):

$$\varepsilon_i^k = \begin{cases} \begin{cases} at_i^k + \frac{C_{p_i}}{SR_{s_k}}, & \text{IMF case} \\ at_i^k + \frac{C_{p_i}}{SR_{p_j}}, & \text{OMF case} \end{cases}, & i \in N \\ at_i^k + \frac{C_{s_k}}{SR_{f_i}}, & i \in F \end{cases}, \quad k \in S,$$

where at_i^k is the adjusted time needed for the preparation and initialisation of the service procedure between the s_k SU and the p_i PU or between the s_k SU and the f_i FU, depending on the application case. The terms SR_{sk}, SR_{pj}, and SR_{fi}, represent the material transfer rate of the equipment corresponding to s_k, p_j and f_i, respectively.

3.2. Arc cost assignment

In the current approach, the arc lengths represent travel times due to the involvement of the service time, ε_i^k, $i \in P$, $k \in S$, that has to be associated with each node.

The cost for the connection between two nodes is affected by the traffic mode that is followed during the operation. Concerning the in-field transport, there are two traffic modes for a SU. According to the first mode, a SU is restricted to manoeuvring only on the field's headlands. As it is shown at Fig. 2a and b, the SU has to execute an 180° turn upon reaching the headland and drive back along the track towards the opposite headland. This regards the case of the implementation of the controlled traffic farming (CTF) system. Moreover, this restriction can be imposed also in the case of uncontrolled traffic farming (UCTF) when according to the current conditions the soil is sensitive to damages. According to the second traffic mode, a SU can freely move in the field body area (Fig. 2c).

In order to build the cost matrix of the VRP, the cost for the connection between each pair of units (SUs, PUs, and FUs) participating in the operation should be determined. Consequently, dedicated path planning methods are needed. Planning methods for AFOs includes developed tools and methods that can be used for the determination of the time and distance that a SU needs to travel between two locations combining both in-field and out-of-the-field transportations (Sørensen *et al.*, 2003; Sørensen and Nielsen, 2005). Bochtis *et al.* (2009a) have developed a discrete event model capable of generating the motion sequence for co-operating machines in material handling operations for both CTF and UCTF systems. For the case of structured agricultural environments, such as orchards, Linker and Blass (2008) presented a method relying on the use of a classic graph search algorithm (A* algorithm) to determine optimal paths, taking into account constraints that are specific to the type of vehicle and the operational environment.

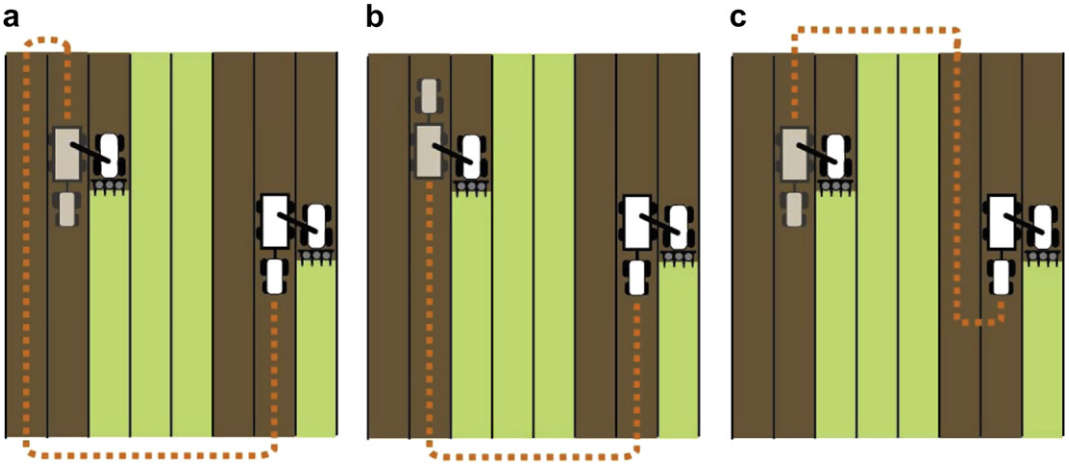


Fig. 2 – a) CTF, on-the-go unloading; b) CTF, stationary unloading; c) UCTF, on-the-go unloading.

The extended graph consists of nodes corresponding to the PUs (customers), to the current position of the SU (virtual "out depot"), and to the FUs (farm depot, stationary trucks, or silo). In problem cases related to deterministic demands, only the cost of paths that connect the PUs and the FUs are considered and it is assumed that they are time-independent. Consequently, the number of the predefined paths is given by

$$\frac{1}{2}\cdot\left(\sum_{i=1}^{|P|} sc_k\right)\cdot\left(\sum_{i=1}^{|P|} sc_k - 1\right) + |F|\cdot\sum_{i=1}^{|P|} sc_k.$$

Fig. 3 depicts the paths that should be determined in the case of one FU and four service calls distributed in different locations in the field area ($\sum_{i=1}^{|P|} sc_k = 4$). Independently of the number of the PUs involved in these calls for the specific example, the total number of the paths equals:

$$\frac{1}{2}\cdot 4\cdot(4-1) + 1\cdot 4 = 10.$$

The interruption of an operation executed by a PU, significantly affects the efficiency of the system as a whole, and thus, it should be avoided. A penalty cost has to be added in order to incur additional cost in a SU when the latest time of a time window is violated. This cost is given by

$$\gamma = \begin{cases} 0 & \text{if } \tau_{ik} \in [a_i, b_i] \\ \kappa(\tau_{ik} - b_i) & \text{if } \tau_{ik} > b_i \end{cases},$$

where κ is a scale factor and τ_{ij} is the total time the SU, s_k, spends before reaches the p_k PU.

3.3. Applications of vehicle routing problem examples to agricultural field operations

3.3.1. Single or multiple SUs in a deterministic IMF operation using a single stationary FU

$$(1 \vee m)\text{-SU}| + \text{IMF}|1 - \text{locFU}| \leftrightarrow \text{VRPTW}.$$

This type of operation refers to a case in which a single SU or a team of identical SUs has/have to refill a single PU or a number of PUs (specifically application units) that operate/s in a deterministic IMF operation (area coverage or orchard operation).

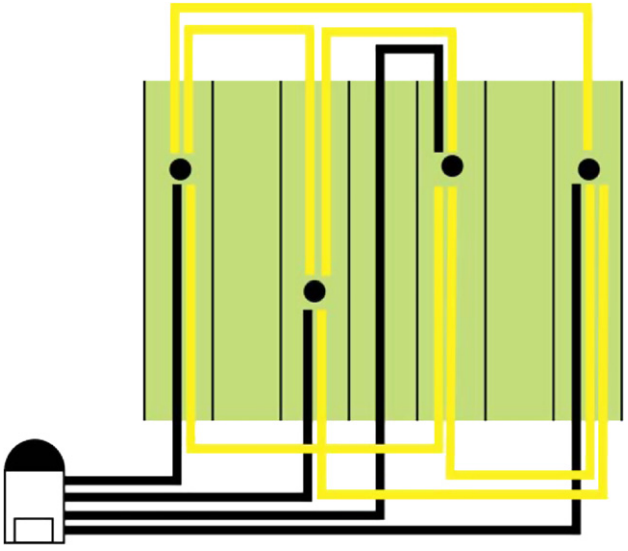


Fig. 3 – Connectivity of the nodes corresponding to the static case of the problem.

In the case of a deterministic material application, the quantity of the material that is applied is predetermined (material per unit area in arable farming or material per tree cluster in orchard farming). Consequently, for a given field pattern followed by a PU, the location where its tank is running empty is known with the uncertainty imposed only by the operator's performance. The time window that is allocated to each service acquisition is relatively short in this case and mainly depends on the tolerance associated with the PU service delays.

3.3.2. Multiple SUs in a deterministic IMF operation using multiple stationary FUs

$$(\equiv \vee \neq)\text{m} - \text{SUs}| + \text{IMF}|\text{m} - \text{locFU}| \leftrightarrow \text{MDVRPTW}.$$

This type of operation differs from the previous one to the fact that there are more than one FU. This case can be cast as a multiple depot VRPTW (MDVRPTW). MDVRPTW is a generalisation of the (single depot) VRPTW, where multiple vehicles start from multiple depots and return to their original depots at the end of their assigned tours.

Methods have been developed also for the case in which vehicles can complete their routes at a different depot from the one that they start from (Cardon *et al.*, 2008), for identical as well as of non-identical vehicles (Dondo *et al.*, 2003).

3.3.3. Single or multiple SUs in a deterministic IMF using a single or multiple mobile FUs

$$(1 \vee m) - \text{SU}| + \text{IMF}|(1 \vee m) - \text{mobFU}| \leftrightarrow \text{VRPTW with } \textit{scheduling horizon}.$$

This type of operation refers to the case in which the FUs are mobile transport units. In this case, a time window, called *scheduling horizon*, should be associated with nodes 0 and $n+1$, *i.e.*, $[a_0, b_0] = [a_{n+1}, b_{n+1}] = [E, L]$, where E and L represent the earliest possible departure from the transport unit and the latest possible arrival at the same unit, as it is imposed by the availability of the transport units.

3.3.4. Single or multiple SUs in a stochastic OMF operation using a single or multiple stationary or mobile FUs

$$(1 \vee m) - \text{SU}| \sim \text{OMF}|(1 \vee m) - (\text{loc} \vee \text{mob})\text{FU}| \leftrightarrow \text{VRPTWSD}.$$

These types of operations refer to the stochastic VRP with time windows (SVRPTW) instance (Changa *et al.*, 2008; Pavone *et al.*, 2009) and specifically to the case of the VRPTW with stochastic demands (VRPTWSD). There are two general methods developed to deal with stochastic optimisation problems; the *a priori* (Bertsimas *et al.*, 1995) and the real-time optimisation methods (Papastavrou, 1996). The first one is based on *a priori* probabilistic information on future or expected events. For the selective harvesting case, for example, the *a priori* information comprise of knowledge of the statistics of the yield spatial

distribution in the field. Such statistics are difficult, if not impossible to obtain. Furthermore, the uncertainty related to the time and location of a "full-grain-tank" event increases the uncertainty of the next similar event, and so on. Due to the dependence between sequential "events", the *a priori* scheduling method is of limited use in AFOs. On the other hand, real-time optimisation methods include the repetitive solution of a new static optimisation problem at a fixed sampling rate based on the latest updated information. For field operations related to unknown demands, real-time optimisation methods are more suitable due to the limited number of PUs (customers) which simplifies planning making re-planning possible in almost real time.

In this type of *dynamic* problem, the paths that connect the current location of each SU with each other location (FUs and PUs) have to be determined each time a re-planning occurs. Consequently, the number of the paths which should be determined is given by

$$\frac{1}{2}\cdot\left(\sum_{i=1}^{|P|} sc_k\right)\cdot\left(\left(\sum_{i=1}^{|P|} sc_k\right)-1\right)+\left(\sum_{i=1}^{|P|} sc_k\right)\cdot|F| +|S|\left(\left(\sum_{i=1}^{|P|} sc_k\right)+|F|\right).$$

Fig. 4 depicts the paths that should be determined in the case of one FU and four service calls distributed in different locations in the field area. The total number of paths equals

$$\frac{1}{2}\cdot 4\cdot(4-1)+4\cdot 1+1\cdot(4+1)=15.$$

Considering the fact that path planning algorithms are highly time-consuming, and a number of paths should be created iteratively for each re-planning of the executed operation, path planning methods with low computational requirements are required.

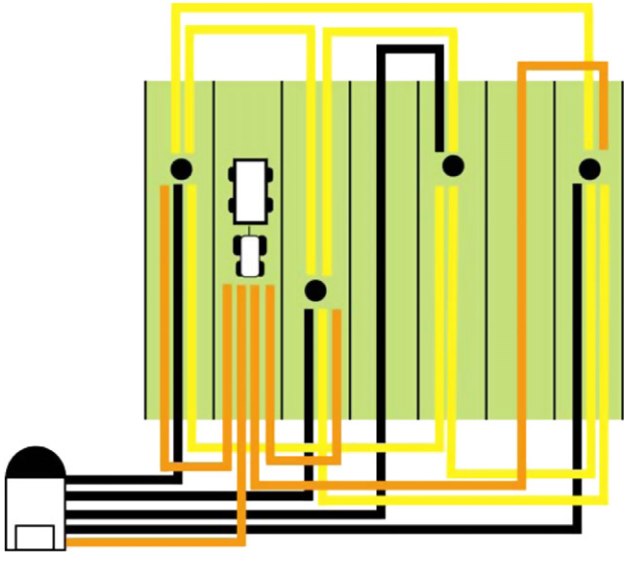


Fig. 4 – Connectivity of the nodes corresponding to the dynamic case of the problem.

3.3.5. *Single or multiple SUs in dynamic IMF or OMF operations using one or multiple stationary FUs*

$(1\vee m)-SU|!(OMF\vee IMF)|(1\vee m)-staFU\leftrightarrow DVRPTW.$

This type of operation refer to the case in which there is no *a priori* information, for example, the case of a refilling unit which supports a spraying unit in a sensor-based spraying operation or a harvesting unit in a selective harvesting operation. This type of operations can be cast as a dynamic vehicle routing problem with time windows (DVRPTW) since the information that is needed for the generation of a set of feasible routes and missions, is dynamically revealed to the decision making system. Consequently, re-planning must be executed as the operation proceeds. There are many methods, from field of dynamic vehicle routing, which can be adopted for the solution of this problem. In terms of simple methods, a new sub-problem is considered each time a new demand appears, or sets of n consecutive demands are formed and deposited in a queue which is served according to a queuing procedure (*e.g.*, the first-come first-served (FCFS) rule, where when a demand appears, it will be served by the first available transport unit). The optimisation procedure considers the number of consecutive demands (n) which constitute a set. In more complicated methods, a spatial grouping of the demand locations can be identified. An example is the modified travelling salesman problem strategy (mod TSP), where for a fixed k, the whole area is divided in k sub-areas (Bertsimas and Van Ryzin, 1993). Within each sub-area, sets are formed with n/k consecutive demands and as they are formed they are deposited in a queue. In more advanced methods, the sub-areas are visited by the vehicle in a given order by passing from one sub-area to an adjacent one. Inside each sub-area, all demands are serviced according to the FCFS rule and this pattern is continuously repeated until all demands are served (Bertsimas and Van Ryzin, 1991).

3.3.6. *Summary of examples*

Based on the identified examples of the VRPTW as related to AFOs, the types of operations are summarised in Table 4.

4. Discussion

In some variants of the VRPTW, the fleet size ($|V|\leftrightarrow|S|$) can be a variable and an additional objective becomes to minimise $|V|$. In these cases, a fixed cost can be added to the cost of all arcs depicting, for example, interest or capital amortisation/depreciation costs. Such optimisation measures can be very useful in the case of sizing and scheduling tasks where the number and the type of the machines have to be determined. Furthermore, the fleet composition can be an issue. Of particular interest in AFOs is the case of mixed fleets composed of a number of different types of machines (although, homogenous in terms of the operation that they execute). For example, a team of refilling units may be composed of tractor-trailer units, as well as, of self-propelled units. For this problems, if $L=\{1,2,\ldots\}$defines the set of the different machine types for each arc $\{i, j\}\in E$, a non-negative three-index cost c_{ij}^{k} should be

Table 4 – VRP examples and their corresponding field operation types

Operation type	VRP example	Field operation examples
$(1 \vee m) - SU \mid + IMF \mid 1 - staFU$	VRPTW	Single or multiple refilling units applied to an application operation and serving a number or application units (*e.g.*, slurry application). All of the refilling units are refilled using a stationary facility, like a storage placed at the farm site
$(\equiv \vee \neq) m - SUs \mid + IMF \mid m - staFU$	MDVRPTW	The IMF operation types employing multiple refilling facilities
$(1 \vee m) - SU \mid + IMF \mid 1 - mobFU$	VRPTW with schedule horizon	The IMF operation types constrained by certain time periods, where a refilling unit can use the FU, for example only when there is a transport truck at the predetermined location providing the necessary load and which also is travelling for refilling
$(1 \vee m) - SU \mid \sim OMF \mid (1 \vee m) - (loc \vee mob) FU \mid$	VRPTWSD	Single or multiple refilling units applied to a harvesting operation with a known *a priori* yield distribution (grain harvesting)
$(1 \vee m) - SU \mid !OMF \mid (1 \vee m) - staFU$	DVRPTW	Selective harvesting denoting the principle of differentiated subfields in terms of harvested produce, for example harvesting with different moisture content or different qualities
$(1 \vee m) - SU \mid !IMF \mid (1 \vee m) - staFU$	DVRPTW	An application unit in a sensor-based variable rate precision spraying without any *a priori* information, for example the an on-line vision based identification of weeds and subsequent control of the spraying unit

assigned for each type $k \in L$. Cases of fixed fleet size (Li *et al.*, 2007), as well as of non-defined fleet size (Choi and Tcha, 2007) are both relevant cases in terms of AFOs. The first case refers to the problem of task assignment to a given machinery team, while the second case refers to the problem of selection of the team for a specific operation.

The work presented here outlines two main aspects/prospects of agricultural logistics. The first one regards the insight into the complexity inherent in AFOs. Since it has been shown that an AFO planning problem can be cast as an instance of VRPTW, it will inhabit the same complexity. The VRPTW is a NP-hard optimisation problem, where even finding a feasible solution to the VRPTW when the number of vehicles is fixed is itself an NP-complete problem[1] (Savelsbergh, 1985). The second aspect regards the potential implementation of existing algorithms adopted from the VRPTW domain. Both scheduling and planning for AFOs can benefit from the development of methods for the solution of the VRPTW instances. A number of exact and heuristic optimisation methods have been developed over recent decades relying on the intelligent exploitation of special problem structures and have benefited from the constant advances in computing capabilities. The theoretical and practical importance of these methods can be further enhanced by constituting the baseline for more complex models for logistic and fleet management in real world applications. The implementation of the resultant fast algorithms could be beneficial for the AFOs, where the interaction of machinery, biological, and environmental conditions generates the need for real-time planning and re-planning.

Advances in information and communication technologies, such as positioning systems, geographic information system (GIS)-based real-time information systems, *Telematic* technologies, monitoring systems for the machinery performance, and sensors for the soil and biomass conditions, can provide the technological basis for the proposed algorithmic-based planning approach to support agricultural logistics and fleet management real-time decisions making and execution controlling.

[1] NP-hard problem (non-deterministic polynomial-time hard) (Papadimitriou and Steiglitz, 1998) is defined because it is not always possible to find optimal solutions to problems of realistic sizes within a reasonable amount of computation time.

5. Conclusions

In this paper, a theoretical approach to the representation of the planning for supporting agricultural machines as instances of the VRPTW has been presented. The approach was motivated by the fact that in agricultural operations involving co-operating machines, a SU, or a team of them, required to fulfil request for on-site service of a PU in a given field region and at a specific time and thus, the PUs can be considered as the ''customers'' in the VRP methodology. It has been shown that scheduling and planning problems for SU in an AFO can be cast as VRPTW instances and, consequently, can be solved using advanced methods developed for the solution of these instances.

Our perspective is that the current approach provides valuable insights for the pursuit of optimal solutions to many current managerial issues on the scheduling and planning for conventional machinery systems, as well as, to future challenging problems related to the implementation of automation and field robotic systems in AFOs.

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